

BAMA 523 Supply Chain Management
Section: BA2

Digital Twin/Beer Game on Steroids

March 23, 2025

Instructor: Dr. Samuel Roscoe

TA: Rajat Singh

Group 6:

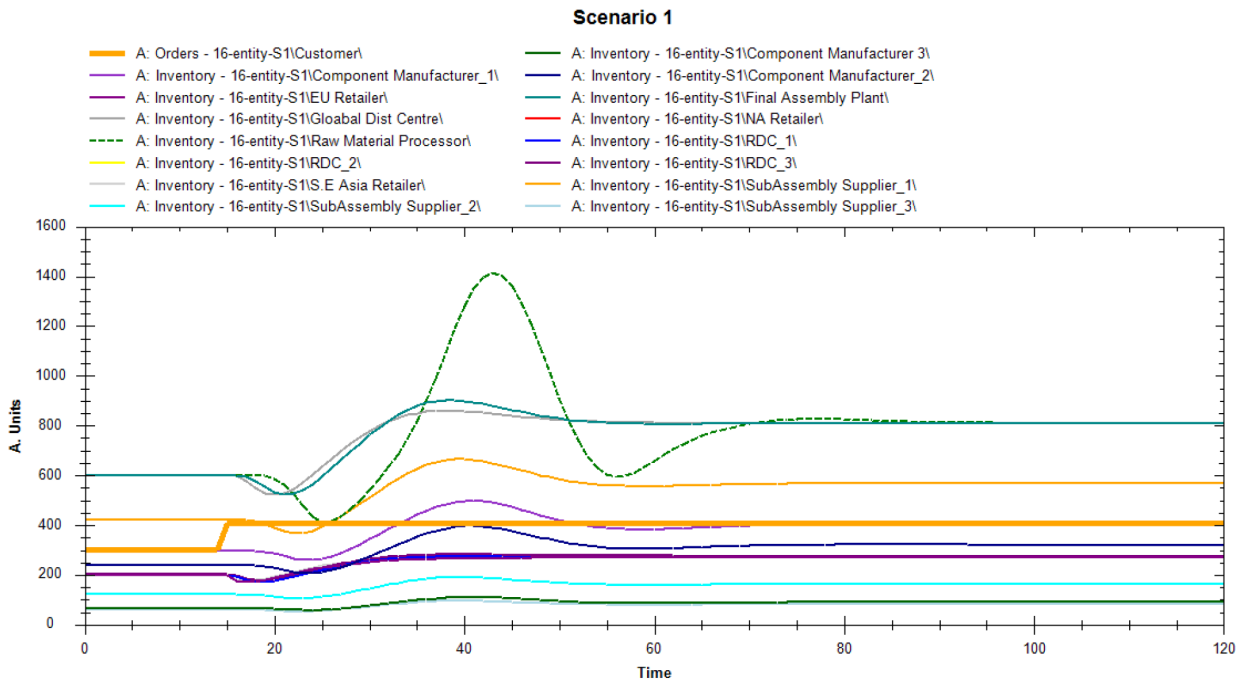
Candice Miao 12978789

Xiaoxi Xu 12491015

Michael Ruizhe Zhang 82889767

Sisly Lyu 72710643

Scenario 1:



In Scenario 1, customer orders were stable at 300 units per week until week 15, when demand increased by 35% to 405 units per week. This led to significant inventory fluctuations across the supply chain as different entities adjusted at varying speeds, creating a bullwhip effect where upstream suppliers experienced amplified demand swings.

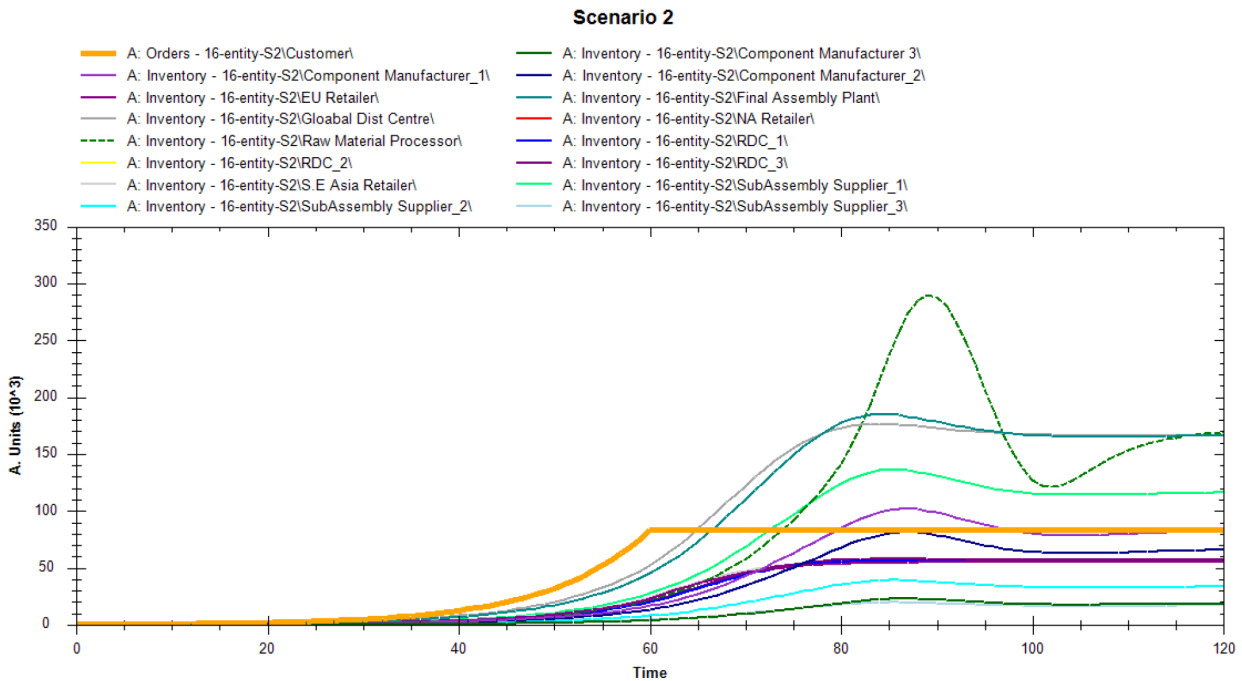
Before week 15, the supply chain was in equilibrium, with retailers, RDCs, manufacturers, and raw material suppliers maintaining stable inventory levels. When demand surged, retailers and RDCs reacted first, increasing their stock to avoid shortages. Their inventory peaked around weeks 20-25 as they ordered more from the Global Distribution Center (GDC), pushing demand further upstream.

As the final assembly plant and GDC responded, inventory levels increased, but due to the 4-week inventory adjustment time, their stock peaked later, around weeks 25-35. This delay caused overproduction at subassembly suppliers and component manufacturers, leading to excessive inventory accumulation.

The Raw Material Processor experienced the most extreme fluctuations, with inventory peaking around week 43 before gradually declining. This suggests a late but exaggerated reaction to increased demand, resulting in overstocking and inefficiencies. Component manufacturers and subassembly suppliers also peaked between weeks 30-45, highlighting the delayed impact of demand changes.

By weeks 40-50, the supply chain stabilized, but overproduction led to excess inventory and inefficiencies. The bullwhip effect was most visible upstream, where poor forecasting caused large stock swings. To prevent this, real-time demand tracking, gradual inventory adjustments, and better coordination between suppliers and manufacturers are essential to reducing disruptions and improving supply chain efficiency.

Scenario 2:



In Scenario 2, customer orders started at 300 units per week and increased by 10% every week from week 2 until week 60. Unlike Scenario 1, where demand spiked suddenly, this scenario featured a gradual increase, leading to continuous inventory growth across the supply chain.

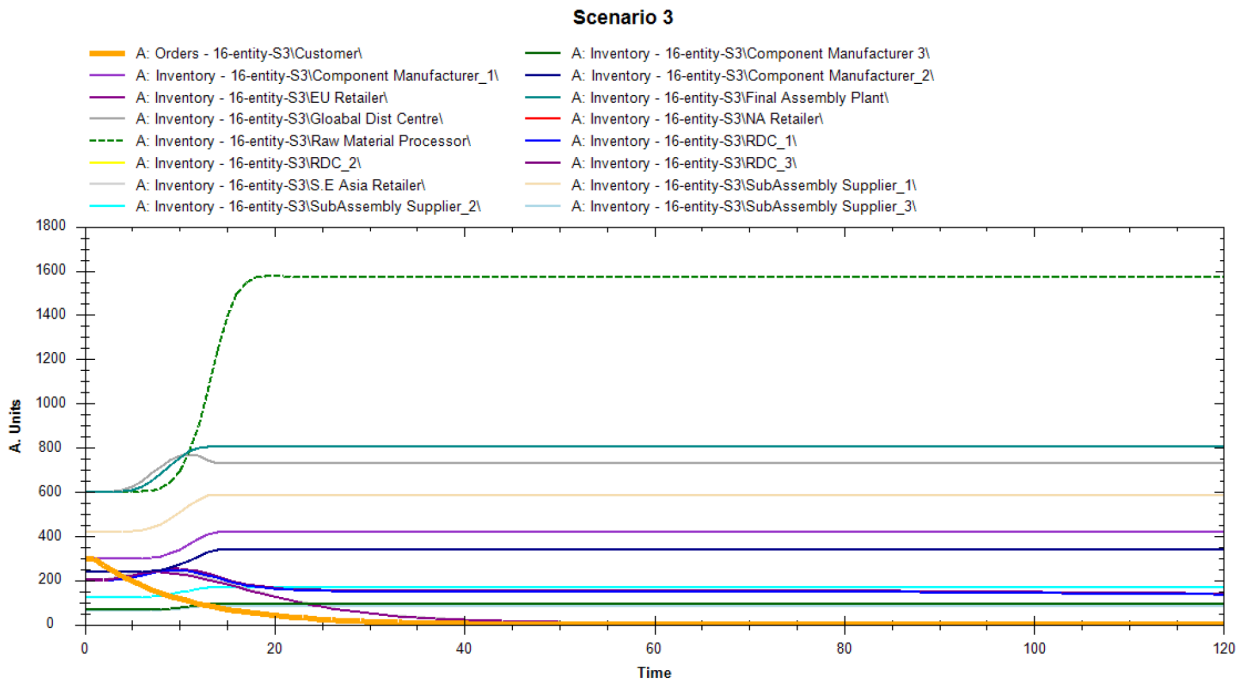
In the early phase (weeks 0-40), retailers and regional distribution centers (RDCs) adjusted inventory gradually to match the rising demand. Since the demand increase was predictable, retailers and RDCs avoided extreme fluctuations and steadily increased stock. However, mid-tier suppliers such as the final assembly plant and global distribution center (GDC) had a delayed response, leading to growing inventory levels as they ramped up production.

Between weeks 40-80, inventory accumulation became more pronounced, particularly in the final assembly plant, subassembly suppliers, and component manufacturers. As orders from downstream entities increased, manufacturers and subassembly suppliers responded with higher production volumes. However, the inventory buildup was not uniform, with some suppliers overproducing, leading to inefficiencies.

The Raw Material Processor exhibited the largest fluctuation, peaking around week 89 before stabilizing. This suggests overcompensation in raw material procurement, causing a temporary stock surplus before adjustments were made. The bullwhip effect was clearly visible, as inventory fluctuations were significantly larger at the upstream levels than at the customer demand level.

After week 80, as demand growth slowed, inventory levels across all entities stabilized. Retailers and RDCs successfully aligned their stock with actual demand, while upstream suppliers continued adjusting to reduce excess inventory.

Scenario 3:



Scenario 3 modeled a continuous 10% weekly decline in customer demand starting in Week 2 and lasting through Week 52. This persistent drop led to a growing disconnect between responsive downstream behavior and sluggish upstream adjustments.

From Weeks 2–10, downstream entities such as retailers and Regional Distribution Centers (RDCs) quickly responded by scaling back orders and reducing inventories. This initial agility helped them avoid overstocking but created a sudden drop in demand signals moving upstream.

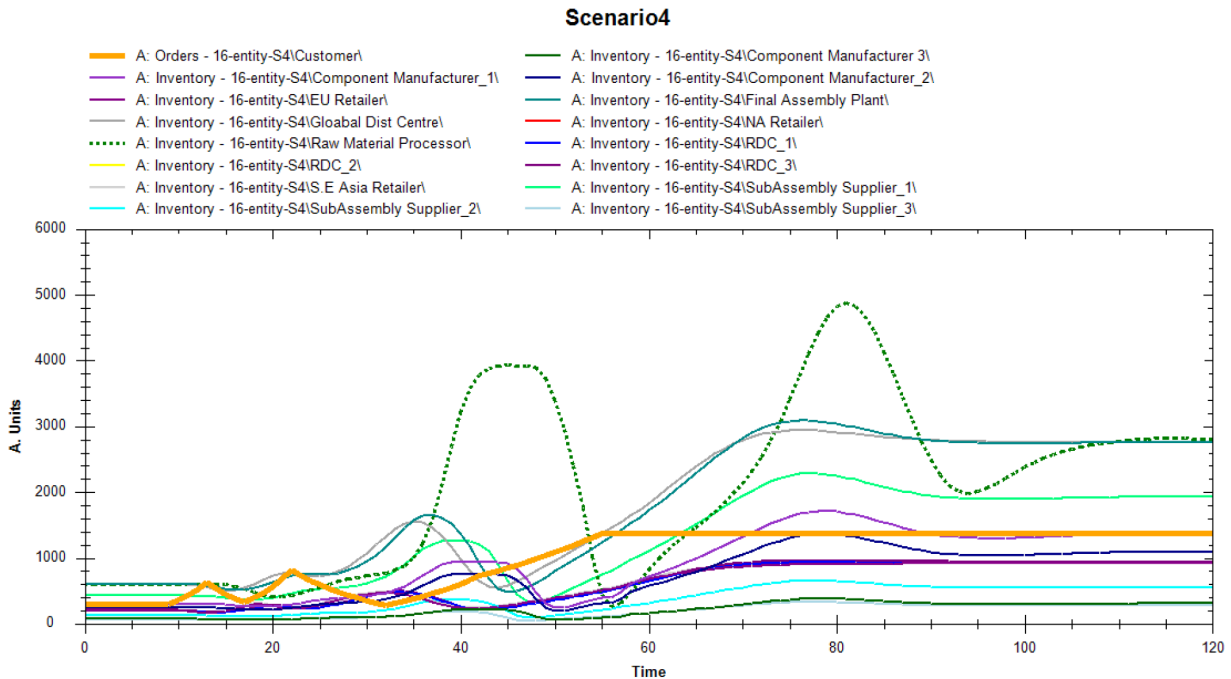
Despite these signals, upstream operations remained largely unchanged. During Weeks 10–25, Final Assembly and Global Distribution Centers (GDCs) continued production at pre-decline levels. This misalignment marked the beginning of a growing mismatch between supply and demand.

By Weeks 25–35, Component Manufacturers and Subassembly suppliers began to experience significant inventory buildup. Production outpaced actual downstream consumption, with excess inventory accumulating rapidly due to lagged responsiveness.

The situation intensified in Weeks 40–50, as the Raw Material Processor peaked with an inventory surplus of around 1,576 units, a clear sign of continued overproduction in the face of shrinking demand.

By Weeks 50–60, demand had dropped to near-zero levels, yet the upstream supply chain remained highly overstocked. This resulted in massive amounts of working capital tied up in unsold inventory, emphasizing the inefficiencies caused by poor demand visibility and inflexible supply strategies.

Scenario 4:



Scenario 4 introduced multiple programmed fluctuations in customer demand, mimicking the unpredictability of real-world markets. These demand shifts included:

- Weeks 10–12: +20% spike (360 units/week)
- Weeks 14–17: -15% drop (306 units/week)
- Weeks 18–22: +20% surge (367 units/week)
- Weeks 23–32: -10% dip (330 units/week)
- Weeks 33–42: +10% increase (363 units/week)
- Weeks 43–55: +5% weekly growth (reaching ~476 units/week)

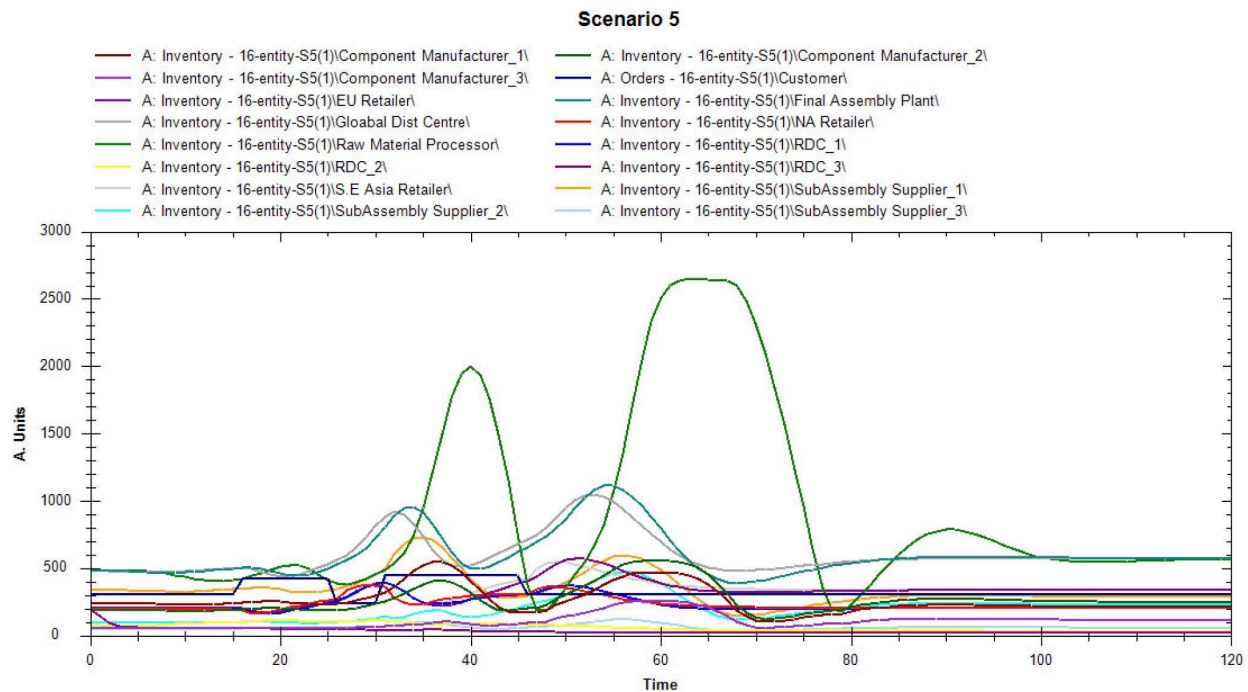
This turbulence triggered sharp inventory and order fluctuations across the supply chain. In Weeks 10–20, retailers and RDCs adapted quickly, adjusting order quantities in near real-time. However, their fast reactions sent volatile demand signals upstream.

By Weeks 25–35, Final Assembly, Subassembly suppliers, and Component Manufacturers started reacting, though often with delays. These lags resulted in uneven inventory levels and supply-demand mismatches.

A major inventory peak occurred around Week 40, followed by another between Weeks 50–55, during which the Raw Material Processor inventory surged to nearly 4,871 units—an extreme response to brief spikes in demand.

As the system moved into Weeks 60–80, market volatility began to settle. The supply chain attempted to recalibrate, but the residual effects of earlier mismatches remained evident across upstream layers.

Scenario 5:



In Scenario 5, the supply chain simulates Zara’s fast-fashion model, where seasonal demand shifts create waves of inventory fluctuations across multiple entities. The simulation reflects several cycles of promotional peaks and low-demand periods, replicating real-world behaviour in the apparel industry.

From Week 1 to Week 15, the supply chain remained relatively stable with consistent customer demand at 300 units per week. Inventory levels across all entities remained balanced with no significant fluctuations.

Between Week 16 and Week 25, a promotional event drove a significant increase in customer demand. This caused inventory levels to rise sharply, particularly in upstream nodes such as component manufacturers, subassembly suppliers, and the final assembly plant. The response across the supply chain was not fully synchronized, leading to a noticeable inventory buildup.

During the Week 26–30 transition phase, demand temporarily dropped. However, inventory levels remained elevated across upstream entities, indicating a lag in production adjustment and overcompensation from previous periods. This delay in response is characteristic of the bullwhip effect.

A second promotional peak from Week 31 to Week 45 again triggered rapid inventory growth. The inventory patterns show repeated amplification in the upper tiers of the supply chain, with slower inventory turnover compared to downstream entities. In contrast, distribution centers and retailers maintained relatively stable stock levels throughout, suggesting that the upstream nodes bore most of the adjustment burden.

By Week 120, customer demand stabilized, and inventory levels across the chain began to normalize. However, the earlier fluctuations highlight inefficiencies in demand forecasting, limited flexibility in production planning, and a lack of real-time coordination between tiers. Overall, the time series reveals the importance of synchronized planning, agile production systems, and demand visibility to mitigate volatility in fast-moving fashion supply chains.